Companion Appendix

Local Income Taxes and Residential Sorting: Municipality- and School-District-Level Aggregations

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This document is the companion appendix to "Local Income Taxes and Residential Sorting: Evidence from Pennsylvania LEHD Origin-Destination Data." The main paper estimates the headline causal effect at the *census-block* level, where both the municipal resident rate and the school-district resident rate enter the regression at their native level. This appendix collects two sets of supplementary results referenced in the main text:

- **Municipality-level aggregations.** The full municipality-level (municipality) Inc/Dec coefficient tables, channel separation, within-design placebo, event-study diagnostics, muni-type heterogeneity, and additional robustness checks. The municipality-level model serves two purposes in the main paper: calibrating the magnitude of the block-level effect against the typical-municipality scale, and recovering the industry decomposition. Both rely on the detailed tables collected here.
- **School-district-level aggregations.** The school-district (SD) is the second rate-setting body in the PA EIT formula. Aggregating the analysis to the SD level produces direction-mixed coefficients that do not survive the realized-dose pre-trend diagnostic. The SD-level results are reported here for completeness; they should be interpreted as

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confirming that the muni-rate effect on residents holds directionally under SD aggregation, not as standalone identification of an SD-channel response.

Section, table, and figure numbering in this appendix is self-contained.

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1 Municipality-Level Full Inc/Dec Detail

Full Inc-cell coefficients with diagnostics

The remainder of this subsection retains the prior detailed municipality-level headline analysis (Inc-cell coefficients with full diagnostics from Equation **??**) for completeness.

Table 1 reports the muni-resident-rate Inc-cell coefficients across the eight outcomes. Each row is a separate regression of the form in Equation (**??**); the displayed columns are the muni-side Inc-bind, Inc-becomes, and Inc-nobind cells. SD and nonresident-rate cell coefficients are estimated jointly in the same regression and are reported in Table **??** (channel separation). Dec-cell coefficients are in Table 2.

Table 1: Headline stacked-continuous DiD: muni-rate Inc cells with diagnostics

Outcome	Inc-bind (n_{muni}=40)	Inc-becomes (n_{muni}=58)	Inc-nobind (placebo) (n_{muni}=26)
Coefficients (per 1 pp of $\Delta\tau^{muni}$)			
log(live_and_work)	-0.402^{***}	-0.181^{**}	-0.062
	(0.096)	(0.072)	(0.169)
log(in_commuters)	-0.213^{**}	-0.197^{***}	-0.187
	(0.083)	(0.054)	(0.204)
log(out_commuters)	-0.053	-0.042	$+0.217^{**}$
	(0.038)	(0.029)	(0.091)
log(total_jobs)	-0.270^{***}	-0.212^{***}	-0.201
	(0.086)	(0.054)	(0.203)
log(total_residents)	-0.100^{***}	-0.068^{*}	$+0.194^{**}$
	(0.033)	(0.040)	(0.095)
share_live_in	-0.045^{***}	-0.012	-0.012
	(0.010)	(0.009)	(0.008)
share_work_outside	$+0.041^{***}$	$+0.020^{***}$	$+0.017$
	(0.016)	(0.007)	(0.014)
Diagnostics (log_live_and_work regression)			
N	218,685		
R^2	0.964		
Within R^2	0.0007		
Dep. var. mean (log_live_and_work)	3.677		
Dep. var. SD	1.934		
Fixed effects	stack-muni + stack-year		
Clusters (muni level)	1,945		

Notes: Stacked continuous DiD, Equation (**??**), drop-incomplete sample with NC controls strictly defined as munis with no rate change on the relevant side. Each muni in a treated cell appears in one sub-experiment (its own first-event cohort); NC munis appear in every sub-experiment. Only muni-side Inc cells are displayed here; SD and nonresident-side cells are estimated jointly in the same regression and reported in Tables **??** and 4. Each regression also includes the share-based control vector (police, streets, debt service / total expenditures; EIT, real estate, intergov / total revenues);

Inc-bind: the binding-cell effect. A 1 percentage-point increase in the muni resident rate in the Inc-bind cell (where the bill rises by the full $\Delta\tau$ for in-muni residents) reduces `live_and_work` by roughly 33 percent (a -0.40 log-point coefficient), `in_commuters` by roughly 19 percent (-0.21 log points), `total_jobs` by roughly 24 percent (-0.27 log points), and `total_residents` by roughly 10 percent (-0.10 log points). The dependent variable mean on `log(live_and_work)` is 3.68, so the typical muni has $e^{3.68} \approx 40$ live-and-work workers; a 1 pp hike at this mean predicts $40 \cdot (e^{-0.402} - 1) \approx -13$ live-and-work workers. Actual rate hikes in the Inc-bind cell average 0.27 pp, which scales the predicted effect down to roughly $0.27 \cdot 40 \cdot (e^{-0.402} - 1) \approx -3.6$ workers per typical event over 5 years.

Inc-becomes: the bill-flip cell. Inc-becomes events show smaller but still significant effects: -0.181 on `live_and_work`, -0.212 on `total_jobs`, -0.068 on `total_residents`. The attenuation relative to Inc-bind is consistent with the bill rising by less than the full $\Delta\tau$ in this cell—only the portion of the rate change above the work nonresident rate is incident on the worker.

Dec cells: thin and noisy. Dec-cell coefficients (Table 2) are reported for completeness but are not robust to specification. Dec-bind has 11 munis; Dec-becomes-non has 4; Dec-nobind has 15. We do not draw asymmetric-response conclusions from these cells.

Table 2: Headline stacked-continuous DiD: muni-rate Dec cells (thin)

Outcome	Dec-bind (n_{muni}=11)	Dec-becomes-non (n_{muni}=4)	Dec-nobind (n_{muni}=15)
log(live_and_work)	$+0.241^{***}$ (0.078)	-0.260 (0.272)	$+0.332^{**}$ (0.165)
log(in_commuters)	$+0.052$ (0.080)	$+0.496$ (0.339)	$+0.013$ (0.121)
log(out_commuters)	-0.068 (0.061)	-0.628 (0.416)	$+0.132$ (0.101)
log(total_jobs)	$+0.111$ (0.073)	$+0.420$ (0.265)	$+0.022$ (0.116)
log(total_residents)	-0.056 (0.043)	-0.651 (0.420)	$+0.135$ (0.099)
share_live_in	$+0.054^{*}$ (0.030)	-0.071 (0.095)	$+0.012$ (0.030)
share_work_outside	-0.011 (0.013)	$+0.014$ (0.017)	-0.003^{*} (0.001)

Notes: Same regression as Table 1. Dec cells are thin and noisy; we do not draw causal conclusions from these coefficients individually. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Per-outcome diagnostics. Table 3 reports N, R^2, within-R^2, and the dependent-variable mean for each headline regression.

Table 3: Headline regressions: per-outcome diagnostics

Outcome	N	R^2	Within R^2	Dep. mean	Dep. SD
log(live_and_work)	218,685	0.964	0.0007	3.677	1.934
log(in_commuters)	239,031	0.959	0.0011	5.847	1.887
log(out_commuters)	242,730	0.979	0.0009	6.636	1.263
log(total_jobs)	239,041	0.961	0.0011	5.948	1.901
log(total_residents)	242,730	0.981	0.0010	6.697	1.299
share_live_in	239,041	0.823	0.0010	0.093	0.076
share_work_outside	242,730	0.945	0.0008	0.943	0.066
share_nonres	239,041	0.823	0.0010	0.907	0.076

Notes: Same regression structure as Table 1. R^2 is the standard within-stack coefficient of determination; within-R^2 is the variation explained by the cell×dose interactions and controls, net of the muni-stack and year-stack fixed effects. Dep. mean and SD computed over the regression sample.

Full main specification with control estimates. Table 4 reports all coefficients from the headline regression on `log(live_and_work)`: muni-side cells, SD-side cells, nonresident-side cells, and control coefficients.

Table 4: Full main spec: log(live_and_work) on all treatments + controls

Term	Coefficient	SE
Muni resident-rate cells (vs NC^{muni})		
Inc-bind $\times\Delta\tau^{\text{muni}}$	-0.402^{***}	0.096
Inc-becomes $\times\Delta\tau^{\text{muni}}$	-0.181^{**}	0.072
Inc-nobind $\times\Delta\tau^{\text{muni}}$	-0.062	0.169
Dec-bind $\times\Delta\tau^{\text{muni}}$	$+0.241^{***}$	0.078
Dec-becomes-non $\times\Delta\tau^{\text{muni}}$	-0.260	0.272
Dec-nobind $\times\Delta\tau^{\text{muni}}$	$+0.332^{**}$	0.165
School district rate cells (vs NC^{SD})		
Inc-bind $\times\Delta\tau^{\text{SD}}$	$+0.177^{*}$	0.099
Inc-becomes $\times\Delta\tau^{\text{SD}}$	$+0.103$	0.090
Dec-bind $\times\Delta\tau^{\text{SD}}$	-0.721^{***}	0.191
Dec-becomes-non $\times\Delta\tau^{\text{SD}}$	-0.223^{**}	0.109
Dec-nobind $\times\Delta\tau^{\text{SD}}$	$+0.048$	0.179
Share-based controls		
share_police	$+0.097$	0.100
share_streets	$+0.049$	0.039
share_debt_svc	-0.058	0.051
share_eit	-0.022	0.085
share_re	-0.182^{**}	0.087
share_intergov	$+0.016$	0.044
Diagnostics		
N	218,685	
R^2	0.964	
Within R^2	0.0007	
Dep. mean (log_live_and_work)	3.677	
Dep. SD	1.934	
Fixed effects	stack-muni + stack-year	
Clusters (muni)	1,945	

Notes: All coefficients from a single regression of log(live_and_work) on cell-specific dose interactions

2 Municipality-Level Heterogeneity by Muni Type

Municipality-Level Heterogeneity by Muni Type

The headline municipality-level coefficient is concentrated in 8 post-industrial PA cities (Wilkes-Barre, Hazleton, Johnstown, Jeannette, Coatesville, Warren, Carbondale, Farrell). Table 5 reports the muni-Inc-bind cell coefficient on `log(live_and_work)` by muni type.

Table 5: Inc-bind cell by muni type (drop sample, post-NC-fix)

Type	n	Mean residents	Inc-bind coef	Sig
City	8	6,904	−0.455	** (p=0.021)
Non-city (borough + township)	32	3,207	−0.194	— (p=0.189)
Borough	10	2,631	−0.213	— (p=0.173)
Township	22	3,468	−0.161	— (p=0.438)
Pooled (headline)	40	—	−0.402***	—

Notes: Stacked-continuous DiD with same controls and FE structure as the municipality-level headline. Each row restricts the sample to its own muni type; NC controls are restricted to the same type. Cluster-robust SEs by muni. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The 8 cities drive the municipality-level headline. The 32 borough and township Inc-bind munis show direction-consistent but individually insignificant effects.

3 School-District-Level Results

School-District-Level Headline Results

Table **??** reports the SD-level dose-response coefficients. The unit of observation is SD-year. The SD's own resident rate enters the regression with binding-cell decomposition; the within-SD aggregate of constituent muni rates enters as a control; and SD-level fiscal share controls constructed from Census F-33 data control for SD fiscal posture. Binding-state

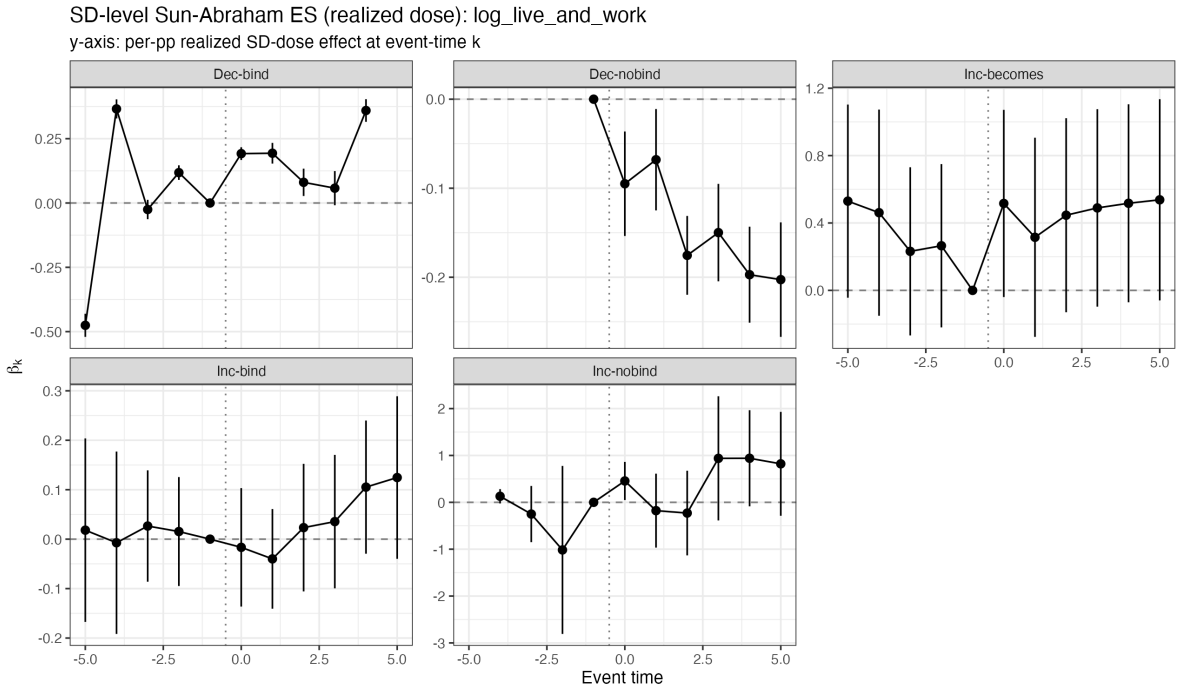
classification at the SD level uses the within-SD mean of `muni_nonres_eit` (the work-side nonresident rate) as the comparator, aggregating the institutional rate field across the SD's constituent munis (the previous draft used the within-SD mean resident rate as a proxy; that proxy has been retired and the cell labels reproduce essentially unchanged under the corrected definition).

Table 6: School-district-level dose-response by binding-state cell

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni	Share work outside
Inc-bind	0.027	0.057	0.106***	0.003	0.000
	(0.038)	(0.046)	(0.030)	(0.004)	(0.004)
Inc-becomes	0.271*	0.184	0.318**	0.004	0.012
	(0.154)	(0.137)	(0.131)	(0.006)	(0.010)
Inc-nobind	0.793	1.149*	0.422***	-0.035	-0.015
	(0.506)	(0.633)	(0.148)	(0.023)	(0.021)
Dec-bind	0.060	0.050	0.178***	0.000	0.006***
	(0.077)	(0.111)	(0.045)	(0.004)	(0.002)
Dec-nobind	-0.126***	0.154***	0.076***	-0.032***	0.021***
	(0.025)	(0.028)	(0.016)	(0.002)	(0.003)
N	51,420	51,454	51,454	51,454	51,454
R^2	0.986	0.987	0.987	0.953	0.977
Within R^2	0.004	0.004	0.005	0.004	0.004
Dep. mean	6.260	8.335	8.769	0.125	0.900
Dep. SD	1.298	1.202	0.909	0.068	0.076

Notes: Cluster-robust standard errors in parentheses. Continuous stacked-DiD with SD-by-stack and year-by-stack fixed effects; standard errors clustered at the school-district level. Unit of observation is SD×year. Binding-cell classification uses the within-SD mean of $\tau^{\text{nr,muni}}$ as the work-side comparator, aggregating the institutional nonresident-rate field over the SD's constituent munis. Rows omitted (no estimable coefficients in any outcome column): Dec-becomes-non. These cells contain too few units, or all coefficients were dropped for collinearity with the stack-interacted fixed effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure 1: SD-level realized-dose ES on `log(live_and_work)` by cell.



Notes: per-pp realized-dose coefficient at each event-time, SD-level stacked-continuous DiD. The SD-resident-rate Inc-bind coefficient is positive in this aggregation, qualitatively different from the muni-level and block-level findings. The pre-event lead is marginally non-zero on `log(live_and_work)` ($\beta_{\text{pre}} = +0.061$, $|t| = 1.74$), and the SD-level results should be read as confirming the directional sign of the muni-rate effect on residents rather than as standalone causal identification of an SD-channel response.

4 Municipality- and SD-Level Pre/Post Act 32 Splits

5 Realized-Dose Event Study Coefficient Detail

This appendix collects the per-cell post/pre ATT summaries of the realized-dose Sun–Abraham event study described in Section **??**. The visual versions appear in the main paper: Figure **??** (pooled, block); Figure **??** (pooled, muni vs SD); Figures **??**–**??** (block decomposition by cell). The tables below report the numerical post-event mean ATT and pre-event lead by binding-state cell.

Table 7: Realized-dose ES — municipality level

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni	Share work outside
Inc-bind (post)	-0.414***	-0.229***	-0.100***	-0.045***	0.047***
	(0.071)	(0.066)	(0.029)	(0.009)	(0.010)
Inc-bind (pre)	0.149***	0.043	0.056	0.016	-0.020***
	(0.056)	(0.061)	(0.059)	(0.012)	(0.005)
Inc-becomes (post)	-0.137***	-0.198***	-0.032**	-0.009**	0.009***
	(0.029)	(0.033)	(0.016)	(0.004)	(0.002)
Inc-becomes (pre)	0.055	-0.006	0.068**	0.004	-0.010**
	(0.043)	(0.083)	(0.029)	(0.008)	(0.004)
Inc-nobind (post)	-0.138***	-0.379***	0.057	0.006	0.017***
	(0.040)	(0.109)	(0.045)	(0.006)	(0.006)
Inc-nobind (pre)	0.169**	0.062	-0.042	0.032**	-0.016***
	(0.070)	(0.127)	(0.148)	(0.013)	(0.006)
Dec-bind (post)	0.650***	0.245**	0.185***	0.012	0.004
	(0.185)	(0.098)	(0.041)	(0.012)	(0.005)
Dec-bind (pre)	0.206	-0.073	0.180***	0.006	0.004
	(0.221)	(0.104)	(0.037)	(0.007)	(0.007)
Dec-becomes-non (post)	1.497***	-0.236***	0.500***	0.127***	-0.024***
	(0.065)	(0.065)	(0.030)	(0.004)	(0.002)
Dec-becomes-non (pre)	0.102	-1.425***	-0.311***	0.086***	-0.022***
	(0.084)	(0.088)	(0.042)	(0.007)	(0.003)
Dec-nobind (post)	-0.328**	-0.006	-0.079	-0.059***	0.021***
	(0.130)	(0.069)	(0.070)	(0.015)	(0.008)
N	198,025	216,226	219,444	216,236	219,444

Notes: Cluster-robust standard errors in parentheses. Each cell reports two rows: post-event ATT (mean of β_k for $k = 0..5$) and pre-event lead (mean of β_k for $k \leq -2$). Rows omitted (no estimable coefficients in any outcome column): Dec-nobind (pre). These cells contain too few units, or all coefficients were dropped for collinearity with the stack-interacted fixed effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 8: Realized-dose ES — school district level

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni	Share work outside
Inc-bind (post)	0.039	0.066***	0.099***	-0.001	0.000
	(0.027)	(0.022)	(0.019)	(0.002)	(0.002)
Inc-bind (pre)	0.013	0.015	-0.015	-0.008	0.001
	(0.039)	(0.031)	(0.025)	(0.005)	(0.003)
Inc-becomes (post)	0.470***	0.311***	0.426***	0.010***	-0.002
	(0.121)	(0.105)	(0.093)	(0.003)	(0.004)
Inc-becomes (pre)	0.372***	0.225*	0.193*	0.013	-0.027***
	(0.139)	(0.119)	(0.105)	(0.009)	(0.009)
Inc-nobind (post)	0.458**	0.934***	0.302***	-0.059***	-0.015
	(0.202)	(0.239)	(0.060)	(0.015)	(0.011)
Inc-nobind (pre)	-0.379	-0.224	-0.171	-0.036**	-0.008
	(0.322)	(0.290)	(0.144)	(0.016)	(0.007)
Dec-bind (post)	0.177***	-0.245***	0.411***	0.062***	0.013***
	(0.011)	(0.011)	(0.006)	(0.001)	(0.001)
Dec-bind (pre)	-0.004	-0.601***	0.200***	0.086***	0.010***
	(0.010)	(0.009)	(0.006)	(0.001)	(0.001)
Dec-nobind (post)	-0.148***	0.147***	0.068***	-0.034***	0.024***
	(0.012)	(0.012)	(0.008)	(0.002)	(0.001)
N	51,128	51,162	51,162	51,162	51,162

Notes: Cluster-robust standard errors in parentheses. Each cell reports two rows: post-event ATT (mean of β_k for $k = 0..5$) and pre-event lead (mean of β_k for $k \leq -2$). Rows omitted (no estimable coefficients in any outcome column): Dec-becomes-non (post), Dec-becomes-non (pre), Dec-nobind (pre). These cells contain too few units, or all coefficients were dropped for collinearity with the stack-interacted fixed effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 9: Realized-dose ES — block level (muni cohort)

Outcome	Log(workers live)	Log(workers work)	Share live+work in muni	Share work outside
Inc-bind (post)	-0.305***	-0.079*	-0.043***	0.043***
	(0.041)	(0.048)	(0.014)	(0.014)
Inc-bind (pre)	-0.027	0.017	0.020**	-0.020**
	(0.053)	(0.033)	(0.008)	(0.008)
Inc-becomes (post)	0.009	-0.046***	-0.030***	0.030***
	(0.020)	(0.015)	(0.006)	(0.006)
Inc-becomes (pre)	0.080***	0.032**	0.003	-0.003
	(0.026)	(0.014)	(0.007)	(0.007)
Inc-nobind (post)	0.045**	-0.050***	-0.042***	—
	(0.018)	(0.019)	(0.008)	
Inc-nobind (pre)	0.123***	0.025	0.008***	—
	(0.019)	(0.016)	(0.003)	
Dec-bind (post)	0.278***	0.294***	-0.052***	—
	(0.055)	(0.110)	(0.010)	
Dec-bind (pre)	0.173***	0.457***	0.028**	—
	(0.057)	(0.099)	(0.011)	
Dec-becomes-non (post)	0.823***	-0.815***	-0.119***	0.119***
	(0.037)	(0.085)	(0.011)	(0.011)
Dec-becomes-non (pre)	0.403***	-1.387***	-0.044***	0.044***
	(0.036)	(0.036)	(0.011)	(0.011)
Dec-nobind (post)	-0.095	0.060	0.015	-0.013
	(36.376)	(0.042)	(11.217)	(11.217)
N	21,404,148	7,988,719	21,404,148	21,404,148

Notes: Cluster-robust standard errors in parentheses. Each cell reports two rows: post-event ATT (mean of β_k for $k = 0..5$) and pre-event lead (mean of β_k for $k \leq -2$). Rows omitted (no estimable coefficients in any outcome column): Dec-nobind (pre). These cells contain too few units, or all coefficients were dropped for collinearity with the stack-interacted fixed effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

6 Binary Sun–Abraham IW Event Study (Municipality and SD)

The binary Sun–Abraham IW event study described in Section **??** complements the realized-dose ES by testing average parallel trends under a binary cohort indicator rather than a continuous treatment-magnitude tag. Block-level binary SA-IW is unstable under stacking (thin cohort $\times$ event-time cells) and is not reported. The tables below report the post-event mean ATT and pre-event lead by binding-state cell at the municipality and school-district levels.

Table 10: Binary Sun–Abraham IW ES — municipality level

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni	Share work outside
Inc-bind (post)	-0.014	0.073***	-0.009	-0.013***	0.003
	(0.030)	(0.027)	(0.015)	(0.003)	(0.003)
Inc-bind (pre)	0.110	0.000	0.054	0.006	-0.012**
	(0.076)	(0.044)	(0.041)	(0.007)	(0.005)
Inc-becomes (post)	-0.134***	0.031	-0.010	-0.012***	0.004**
	(0.021)	(0.022)	(0.008)	(0.003)	(0.002)
Inc-becomes (pre)	-0.114***	0.064	0.058***	-0.027***	-0.002
	(0.033)	(0.042)	(0.010)	(0.004)	(0.004)
Inc-nobind (post)	0.180**	0.091	0.021	0.013***	-0.004
	(0.078)	(0.102)	(0.020)	(0.005)	(0.005)
Inc-nobind (pre)	0.113	-0.300**	-0.079***	0.011*	-0.002
	(0.077)	(0.130)	(0.016)	(0.006)	(0.005)
Dec-bind (post)	-0.191***	-0.305***	-0.020	-0.011*	-0.002
	(0.062)	(0.067)	(0.028)	(0.006)	(0.005)
Dec-bind (pre)	-0.253***	-0.038	-0.187***	-0.007*	0.009
	(0.087)	(0.061)	(0.026)	(0.004)	(0.006)
Dec-becomes-non (post)	0.046	0.055	0.010	0.002	0.041
Dec-becomes-non (pre)	0.865	0.140	-0.087	0.058	-0.036
Dec-nobind (post)	-0.388***	-0.393***	-0.057**	0.018**	0.021***
	(0.089)	(0.071)	(0.025)	(0.009)	(0.008)
Dec-nobind (pre)	0.788***	0.174***	0.228***	0.058***	-0.010**
	(0.048)	(0.037)	(0.019)	(0.004)	(0.004)

Notes: Cluster-robust standard errors in parentheses. Binary Sun–Abraham interaction-weighted ES. Post = mean of β_k for $k = 0..5$; pre = mean for $k \leq -2$. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 11: Binary Sun–Abraham IW ES — school district level

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni	Share work outside
Inc-bind (post)	0.058*	0.030	0.097***	-0.001	0.005*
	(0.031)	(0.024)	(0.023)	(0.002)	(0.003)
Inc-bind (pre)	0.099***	0.110***	0.079***	-0.003	-0.014***
	(0.024)	(0.027)	(0.016)	(0.002)	(0.002)
Inc-becomes (post)	1.226***	0.984***	1.078***	0.038***	-0.017***
	(0.141)	(0.112)	(0.112)	(0.004)	(0.002)
Inc-becomes (pre)	1.111***	1.505***	0.679**	-0.055***	-0.039***
	(0.417)	(0.345)	(0.341)	(0.011)	(0.005)
Dec-bind (post)	0.074	0.035	-0.009	0.004	-0.011
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Dec-bind (pre)	-0.174	0.077	-0.129	-0.028	0.010
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)

Notes: Cluster-robust standard errors in parentheses. Binary Sun–Abraham interaction-weighted ES. Post = mean of β_k for $k = 0..5$; pre = mean for $k \leq -2$. Rows omitted (no estimable coefficients in any outcome column): Inc-nobind (post), Inc-nobind (pre), Dec-becomes-non (post), Dec-becomes-non (pre), Dec-nobind (post), Dec-nobind (pre). These cells contain too few units, or all coefficients were dropped for collinearity with the stack-interacted fixed effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

7 Dose-Magnitude Linearity (Municipality, SD, Block Terciles)

8 Alternative Control Group: Clean-in-Window

9 Data Construction Pipeline (Step-by-Step)

This appendix expands the build pipeline summarized in the main paper's Data section. The analysis links two administrative systems that do not share common identifiers: LEHD/LODES employment records (Census block geography) and Pennsylvania EIT rates (DCED municipality IDs). The build proceeds through the following steps.

1. **Scrape PA DCED MunStats tax rates.** PA does not publish a downloadable panel of historical EIT rates. The only source is the DCED MunStats portal, which serves one XLS file (∼12 MB) per year via an ASP.NET form. A Python script submits requests for each year 2000–2026, extracting resident and nonresident EIT rates, school district EIT rates, and Local Services Tax for all ∼2,500 municipalities. About 27% of munis report multiple school-district rates; in the municipality- level panel each municipality-year is assigned the maximum SD rate.

2. **Scrape PA DCED Annual Financial Reports.** Fiscal controls (share-based) are constructed from the DCED AFR (71 columns per municipality-year) for 2002–2023, trimmed at load to 2002–2019: shares of police, streets, debt service in total expenditures and shares of EIT, real estate tax, intergovernmental revenue in total revenues.

3. **Backfill nonresident rates.** The DCED nonresident registry was incomplete pre-Act 32: 55–60% coverage 2002–2011, 72% in 2012, 77% by 2019. Diagnostics show this is a reporting gap, not a regime gap (0.2% of late-reporting munis have a true zero nonresident rate; 96.3% of nonresident rates are perfectly stable across 2012–2019). I backfill missing pre-2012 nonresident rates with the earliest observed post-2012 rate per muni and flag the imputation.

4. **Download LODES block-level OD files.** LODES Version 8 (2020 Census geography retroactively applied to 2002–2021), block-level (`agg_geo = "block"` in the `lehdr` R package). I download main files (intra-state) and auxiliary files (cross-state). All-jobs (S000), earnings (SE01, SE03), and age (SA01–SA03) segments. Reference date: April 1.

5. **Crosswalk PA DCED to Census geography.** DCED municipality names matched to TIGER/Line COUSUBFP via normalized name + county; 99.6% match (9 unmatched munis dissolved or merged before 2020).

6. **Build a block-to-municipality crosswalk.** Point-in-polygon assignment of TIGER 2020 block centroids to PA cousub geometries. Because 2020 Census blocks nest in cousubs by construction, this assignment is exact: every block has an unambiguous municipality. The crosswalk recovers all 2,561 PA municipalities. The alternative (tract-centroid assignment) silently drops 1,082 of 2,561 municipalities (42%) and misattributes employment for the retained ones.

7. **Compute neighbor pairs.** Centroid-to-centroid distances between all PA muni pairs; retain pairs within 80 km.

8. **Aggregate LODES.** Block-pair-year LODES is summed to (work municipality, home municipality) × year, producing total jobs, live-and-work, in-commuters, and cross-state commuters per municipality-year. Block-level outcomes (`home_count`, `work_count`, `home_in_muni_share`) are retained directly on the block-year panel without aggregation.

9. **Merge rates and controls.** Final join: rate panels and fiscal controls onto the OD panels, producing the analysis-ready municipality × year, school- district × year, and block × year datasets.

10 Pooled Inc/Dec Binary-Event Analog

The binary-event analog of the per-pp pooled specification (Equation **??**) replaces each $\Delta\tau_{it}^{s}$ dose interaction with a post-treatment indicator that equals one in the event year and after. The four reported coefficients per regression are $\{\text{everInc}^{M}, \text{everDec}^{M}, \text{everInc}^{SD}, \text{everDec}^{SD}\}$. Sign-wise the binary estimates agree with the per-pp pooled headline at every level.

Table 12: Pooled Inc/Dec headline — block level (binary event)

Outcome	Log(workers live)	Log(workers work)	Share live+work in muni	Share work outside
Municipality rate (τ^M)				
$\Delta\tau > 0$ (Inc)	-0.052**	-0.038**	-0.018**	0.018**
	(0.025)	(0.016)	(0.008)	(0.008)
$\Delta\tau < 0$ (Dec)	-0.050***	-0.028	0.013***	-0.013***
	(0.017)	(0.053)	(0.004)	(0.004)
School district rate (τ^{SD})				
$\Delta\tau > 0$ (Inc)	0.003	-0.000	0.001	-0.001
	(0.006)	(0.007)	(0.001)	(0.001)
$\Delta\tau < 0$ (Dec)	0.008	0.066***	0.005***	-0.005***
	(0.006)	(0.015)	(0.001)	(0.001)
N	23,567,655	8,751,581	23,567,655	23,567,655
R^2	0.858	0.865	0.563	0.563
Within R^2	0.000	0.000	0.000	0.000
Dep. mean	2.238	2.318	0.121	0.879
Dep. SD	1.186	1.655	0.184	0.184

Notes: Binary post-treatment indicator: 1 in the event year and after. Cluster-robust standard errors in parentheses. Each regression includes both rate-setters' direction-pooled treatments jointly; the four reported coefficients are β^M_{Inc}, β^M_{Dec}, β^{SD}_{Inc}, β^{SD}_{Dec}. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 13: Pooled Inc/Dec headline — municipality level (binary event)

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni	Share work outside
Municipality rate (τ^M)					
$\Delta\tau > 0$ (Inc)	-0.117***	-0.099***	-0.000	-0.012***	0.014***
	(0.034)	(0.034)	(0.020)	(0.004)	(0.003)
$\Delta\tau < 0$ (Dec)	-0.141*	-0.049	0.038	-0.010**	0.000
	(0.074)	(0.038)	(0.058)	(0.005)	(0.002)
School district rate (τ^{SD})					
$\Delta\tau > 0$ (Inc)	0.031	0.062**	0.015***	0.001	-0.001
	(0.020)	(0.026)	(0.006)	(0.001)	(0.001)
$\Delta\tau < 0$ (Dec)	-0.007	0.023	-0.013	-0.002	-0.003
	(0.078)	(0.114)	(0.016)	(0.003)	(0.002)
N	218,677	239,023	242,722	239,033	242,722
R^2	0.964	0.959	0.981	0.823	0.945
Within R^2	0.001	0.002	0.001	0.001	0.001
Dep. mean	3.677	5.847	6.697	0.093	0.943
Dep. SD	1.934	1.887	1.299	0.076	0.066

Notes: Binary post-treatment indicator: 1 in the event year and after. Cluster-robust standard errors in parentheses. Each regression includes both rate-setters' direction-pooled treatments jointly; the four reported coefficients are β^M_{Inc}, β^M_{Dec}, β^{SD}_{Inc}, β^{SD}_{Dec}. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 14: Pooled Inc/Dec headline — school district level (binary event)

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni	Share work outside
School district rate (τ^{SD})					
$\Delta\tau > 0$ (Inc)	0.089*	0.091**	0.129***	0.000	0.005
	(0.051)	(0.046)	(0.035)	(0.003)	(0.004)
$\Delta\tau < 0$ (Dec)	-0.007	-0.072***	-0.069***	0.008*	-0.006***
	(0.042)	(0.020)	(0.023)	(0.004)	(0.002)
N	51,440	51,474	51,474	51,474	51,474
R^2	0.986	0.987	0.987	0.953	0.977
Within R^2	0.004	0.007	0.006	0.006	0.005
Dep. mean	6.260	8.335	8.768	0.125	0.900
Dep. SD	1.298	1.202	0.909	0.068	0.076

Notes: Binary post-treatment indicator: 1 in the event year and after. Cluster-robust standard errors in parentheses. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

11 Sample Variant Robustness

References